

FORCE AND PRESSURE

Have you ever experienced pressing a pencil between your fingers?

What did you feel on the finger at the tail of pencil?

What did you feel on finger at the tip of the pencil?

On which side did you feel more pressure?

While you put same force on both sides why did you feel different effect on the fingers?

Can you repeat this activity with a new pencil before sharpening it?

Repeat the activity with a pencil sharpened on both sides.

Feel the different effect of force and discuss the cause of effect in each case.

In this Chapter you will learn about:

- Pressure, Force and Area.
- Units (N/m^2 and Pascal).
- Hydraulics and Hydraulic Systems.
- Water Pressure.
- Pneumatics (How Gases Behave under Pressure?).
- Gas Pressure in a Container
- Aerosols.
- Atmospheric Pressure.

All the students will be able to:

- Define the term Pressure.
- Identify the units of Pressure.
- Explain hydraulics and hydraulic system by giving examples.
- Explain how gases behave under pressure.
- Describe the causes of gas pressure in a container.
- Explain the working of Aerosols.
- Identify the application of gas pressure.
- Describe the term atmospheric pressure.



Fig.7.1 (a)



Fig.7.1 (b)



Fig.7.1 (c)

PRESSURE, FORCE AND AREA:

➤ Define the term Pressure

Rani and Qaima visited Thar in summer vacation. One day they decided to go for walk on sand mounds. Rani put on her favourite high heel sandals, meanwhile Qaima put on her shoes with flat bottom.

As soon as they stepped out onto the sand, Rani's heels sink right through while Qaima has no trouble walking on sand in her flat



Figure 7.2



bottom shoes! Why did this happen? If Rani and Qaima had almost the same weight and the same size, why did Rani fall through the sand while Qaima did not?

This occurred because even though both girls exerted the same force on the ground, Qaima's shoes spread that force over a larger area, causing the pressure on the ground to be much less.

Pressure describes how a force is spread over an area. It is a measure of force acting on a certain area. Pressure is calculated by the following formula;

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

You can notice that the force is applied in a direction that is perpendicular to the surface of the object.

Pressure depends on two factors

1. Forces acting
2. Area of surface

Formula shows that larger the force, greater will be its pressure. Similarly, the smaller the surface area, greater will be the pressure.

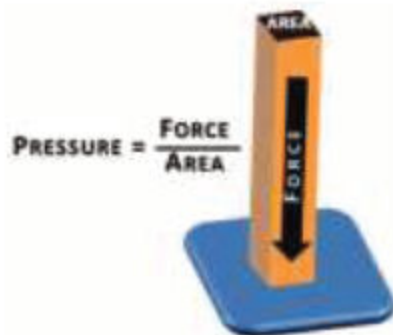


Fig. 7.3 Force acting to a surface

Example 1:

A hammer is used to drive a nail into a wooden floor. The hammer is brought down with a force of **200 N**. The area of the top of the nail is **0.5 cm²**. What pressure is put upon the top of the nail by the hammer blow?



Fig.7.4

Solution:

$$\text{Force} = F = 200\text{N}$$

$$\text{Area of nail} = A = 0.5 \text{ cm}^2$$

$$\text{Pressure} = P = ?$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{F}{A}$$

$$P = \frac{200\text{N}}{0.5\text{cm}^2} = \mathbf{400 \text{ N/cm}^2}$$

Example 2:

A truck has tyres of area 5 m^2 . It puts a pressure of 15000 N/m^2 on the road. What is the weight of the truck?

Solution:

Given: Area = $A = 5 \text{ m}^2$

$$\text{Pressure} = P = 15000 \text{ N/m}^2$$

$$\text{Weight} = W = F = ?$$

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{F}{A} \quad \text{—————} \quad F=W= P \times A = 15000 \times 5$$

$$\mathbf{W = 75000\text{N}}$$

DO YOU KNOW?

Pressure examples in daily life :

1) School bags have wider straps :



2) Sharp knife cuts better than blunt knife:



Blunt knife: more area
Sharp knife: less area

3) Buildings have wider foundation:



Broader foundation
less pressure on ground

4) Easier to walk on sand with flat shoes rather than shoes with sharp heels:



Activity 7.1:

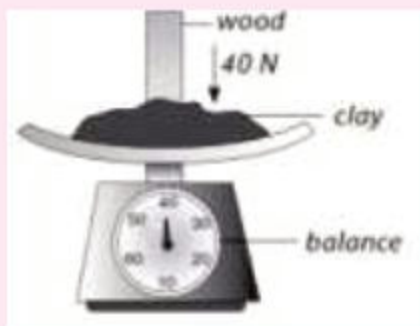
Effect of Area and Force on Pressure

Materials needed:

Compression spring balance or electronic top-pan balance, metre ruler, lump of clay, piece of wood 20 cm x 1.5 cm x 1.5 cm, piece of wood 20 cm x 4.5 cm x 4.5 cm, pencil

Procedure:

- Place the lump of clay on the pan of balance as shown in the diagram below.
- Take the piece of wood of cross-section 1.5 cm x 1.5 cm and push it down into the clay until the scale reads 40 N. Make a pencil mark on the wood to show how deep it has gone into the clay. Then remove the piece of wood.
- Measure how high the pencil mark is from the end of the piece of wood. This will tell you the depth of the dent in the clay.
- Repeat stages 2 and 3 above, but this time using the piece of wood of cross-section 4.5 cm x 4.5 cm. Push it down into a new area of the clay.
- Record your measurements in the table below:



Cross section of Wood	Depth of the dent (cm)	Force, F (N)	Area, A (cm ²)	Pressure, P = Force/Area (N/cm ²)
1.5 cm x 1.5 cm				
4.5 cm x 4.5 cm				

a) Which piece of wood has made the deeper dent?

b) Which piece of wood has exerted the greater pressure?

c) Why the same force has exerted a different pressure on the clay?

d) Assume the two pieces of wood as the heels of shoes, which would cause the most damage to a floor?

UNITS OF PRESSURE

➤ Identify the units of Pressure

In Pressure, force is expressed in Newtons (N) and area is expressed in square meters (m^2). Therefore, pressure is expressed in N/m^2 , which is the SI unit for pressure. This unit is also called the **Pascal** (abbreviated Pa). As Pascal is a very small pressure, therefore pressure may also be expressed in the kilopascal (kPa), which equals 1000 Pascals. For example, the correct air pressure inside a bike tyre is usually about 200 kPa (30psi).

Blaise Pascal was a French Scientist whose discoveries about pressure in fluids led to pascal's law after his name.

WATER PRESSURE

The amount of pressure at your tap can depend on how much high the service reservoir or water tower is above your home, or on how much water other consumers are using. The height of your house can also affect water pressure, houses at the top of a hill may receive lower pressure than houses which are at the bottom of the hill. The amount of force per unit area that water exerts, say, on the container in which it is or on a submerged object is called water pressure. The pressure of water that is not flowing depends on the depth. Pressure is the force that pushes water through pipes. Water pressure determines the flow of water from the tap.

Activity 7.2:

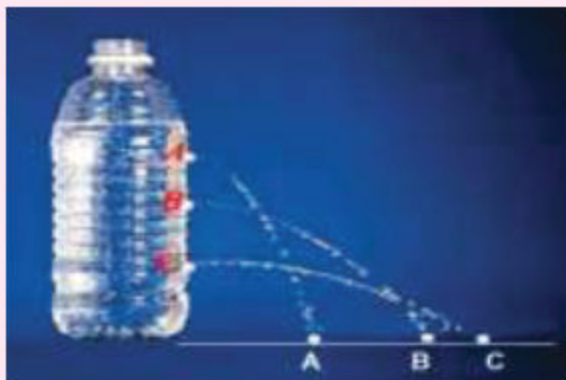
Water pressure at Different Heights

Materials:

A plastic bottle or other available container of Water, Tool to drill at least three holes

Activity Setup:

Make 3 holes (A, B and C) in the bottle in a vertical arrangement. Place the holes at different heights with at least 1 inch apart.



Procedure:

- Cover the holes with your fingers
- Fill the bottle with water and place the cap on the bottle.
- Expose the holes and observe the water flow from each hole.

Note: You can quantify the activity by using a ruler to measure the initial contact points of the water flow from each hole.

Questions:

1. Observe the flow of water through the holes. What are your initial observations?

2. What appears to happen to the flow over time?

3. Explain why there are variations in the length of the water discharge from each of the three holes.

HYDRAULICS AND HYDRAULIC SYSTEMS - PASCAL'S LAW

- Explain Hydraulics and Hydraulic System by giving examples.

Pascal's law states that when a liquid is placed in a container, the pressure applied to the fluid by the container is equal throughout the container. This phenomenon is used for hydraulic devices.

Hydraulics:

Hydraulics is a branch of science that deals with practical applications of liquid in motion. The pressure in a liquid is transmitted equally in all directions, so a force exerted at one point on a liquid will be transmitted to other points in the liquid. This technology is called Hydraulics. You must have seen the water

pistols, car lifters and big cranes. All these objects are using power of moving fluid.

Activity 7.3:

Pressure Transmitted Equally in All Directions

Materials:

1 balloon, water, needle or something to poke holes in balloon with.

Setup:

Fill balloon with water

Procedure:

- Tie off balloon so there is very little air trapped inside.
- Quickly poke several holes around the perimeter of the balloon and squeeze.
- Observe the flow of water through the holes in the balloon.

Questions:

1. What are your initial observations?

2. How would you describe the flow of water from hole? Explain your observations.



Hydraulic Systems:

Pressure can be transmitted through liquids. In hydraulic systems, exerting a small force over a small cross-sectional area can lead to pressure being transmitted, creating a large force over a large cross-sectional area. This ability to multiply the magnitude of forces allows hydraulics to be used in many applications such as car-braking systems. Hydraulic systems are also found in:

- lifting equipment example: hydraulic jacks and swing chair lifts.

- lifting and excavating machinery such as diggers.
- hydraulic presses which are used during the forging of metal parts.
- wing flaps and some rudders on aircraft and boats.

1. Car Braking System:

Car braking system cause a relatively small force from the driver's foot to produce a greater force, which acts equally on all four brake pads. In a car brake system the force from the driver's foot exerts pressure on the brake fluid in a small piston.

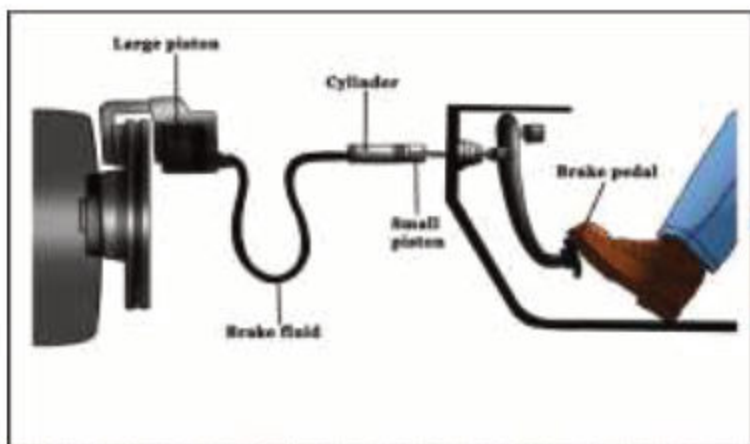


Fig. 7.4 Car Braking System

The brake fluid squirts down a long, thin pipe until it reaches another cylinder at the wheel which is much wider. When the fluid pushes the piston of greater cross section area, the transmitted pressure acts on this larger area to produce a larger load force on the brake pads. The pads then rub against the brake discs and cause the car to slow down.

2. Hydraulic Jack System:

A hydraulic jack is a device used to lift heavy loads. The device itself is light, compact and portable, but capable of exerting great force. The device pushes liquid against a piston, pressure is built in the jack container. The jack is based on Pascal's law that the pressure of a liquid in a container is the same at all points.

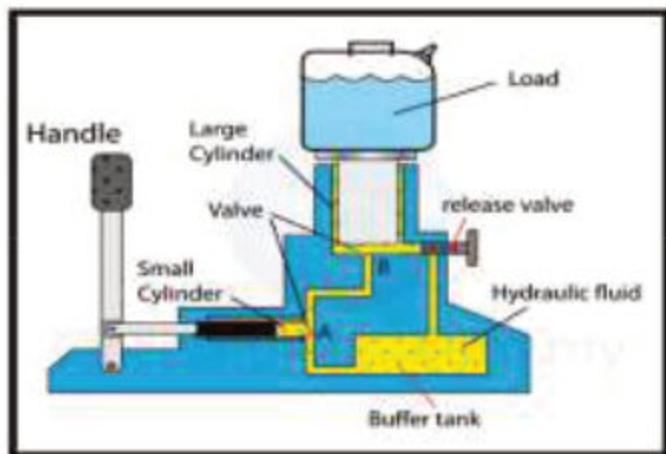


Fig. 7.5 Hydraulic Jack

How Hydraulic Jacks Work When the handle is pressed down, valve A is closed whereas valve B is opened. The hydraulic fluid is forced into the large cylinder and hence pushes the piston moving upward. When the handle is raised, valve B will be closed while valve A will be opened. Hydraulic fluid from the buffer tank will be sucked into the small cylinder. This process is repeated until the load is sufficiently lifted. The large piston can be lowered down by releasing the hydraulic fluid back to the buffer tank through the release valve. Essentially, if two cylinders (a large and a small one) are connected and force is applied to one cylinder, equal pressure is generated in both cylinders. However, because one cylinder has a larger area, the force the larger cylinder produces will be higher, although the pressure in the two cylinders will remain the same. Hydraulic jacks depend on this basic principle to lift heavy loads.

GAS PRESSURE IN A CONTAINER

- Describe the causes of gas pressure in a container

Gas pressure in a closed container is the result of the gas molecules hitting the inside wall of the container. The molecules move about and are attempting to escape the container. When they cannot escape, they strike the inside wall and then bounce around. The more molecules striking the inside wall of the container, the greater the pressure. This concept represents the kinetic theory of gases.

• Turning Up the Heat

Changing the temperature affects pressure in a closed container. Raise the temperature, and the pressure increases. This occurs due to the increased movement of the gas molecules. Double the temperature and you double the pressure.

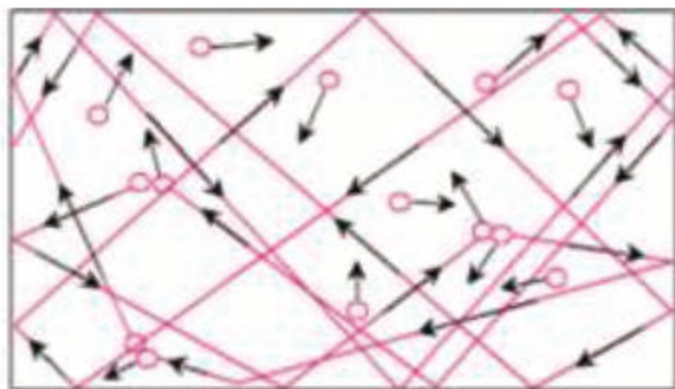


Fig. 7.6 The motion of gas molecules in a closed container

• More Pressure, Lower Volume

The volume of a gas and its pressure are inversely related. Decrease the volume, and the pressure increases. As the volume a gas occupies decreases, the molecules of the gas are forced closer together, but their movement continues. They have less distance to travel to impact the container walls, so they strike more often, thus creating more pressure.

• Density of the Gas

Increase the number of particles in a container, and the pressure of the system within the container increases. More molecules mean more hits against the container walls. Increasing the number of particles means you have increased the density of the gas.

PNEUMATICS

- Explain how gases behave under pressure

Pneumatics is an aspect of science and engineering that is concerned with using the energy in compressed gas to make something move or work. The origins of pneumatics trace back to the first century when the Greek mathematician Hero of Alexandria created mechanical systems powered by wind and steam and documented his processes. Today, pneumatics plays an important role in manufacturing. Pneumatic systems are similar to hydraulics in function, but hydraulic systems use liquid instead of gas.

DO YOU KNOW?

The word Pneumatics comes from the Greek word *pneuma*, which means 'breath or wind'

Applications of Pneumatic Systems

- Identify the applications of gas pressure

The use of pneumatic Systems is not restricted to factories. You just need to have a look around and you will see them everywhere. In our daily life, we use many items in which pneumatic systems

are present.

A few common examples of things we use in our daily life that contain pneumatic Systems are:

1. Spray Gun:

Spray gun, a painting tools use compressed air from a nozzle to atomize a liquid into a controlled pattern. The spray nozzle operates by the effect of high-velocity turbulent air on the surface of filaments or films of liquid, causing them to collapse to droplets with a wide range of sizes.



Fig. 7.7 Spray Gun

2. Bicycle Pump:

A *bicycle pump* is essential for keeping your tyres inflated. The body of the bicycle pump is a cylinder and the handle is a plunger shaped to form a non-return valve in the cylinder. When you pull the pump handle out the plunger allows air to fill the cylinder and pushing the handle into the pump compresses the air until it is at a higher pressure than the air in the bicycle tyre. The valve in the tyre allows air into the tyre and closes when the pressure drops. Each time the pump is pumped a little more air goes into the tyre until the tyre is at the correct pressure for riding.



Fig. 7.8 Bicycle Pump

3. Vacuum Cleaner:

A vacuum cleaner is a device that uses an air pump to create a partial vacuum to suck up dust and dirt, usually from floors. Compressed air vacuum cleaner is powerful, multi-functional and easy to handle.

4. Dental Drill:

The dental drill is a tool used by dentists to bore through tooth

enamel as well as to clean and remove plaque from the tooth's surface. It is composed primarily of a hand piece, an air turbine, and a tungsten carbide drill bit. The modern dental drill has enabled dentists to work more quickly and accurately than ever before, with less pain for the patient.



Fig. 7.9 Vacuum Cleaner



Fig. 7.10 Dental Drill

Activity 7.4:

Controlling movement using a syringe attached to each end of plastic tubing

Materials:

Two syringes of same size one syringe of different size,

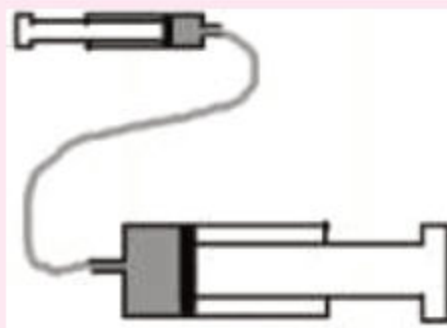
Plastic Tubings

Procedure:

a) Using 2 syringes of the same size:

Push the end of one syringe fully in and attach the tubing to it.

Push the end of the other syringe partly in and attach the tubing to it. Make sure that the syringes are not pushed out of the tubing.



Questions:

1. Predict what will happen to the other syringe when you push one syringe in and out?

2. Why does this happen?

3. Can you compare the distances both syringes move?

b) Repeat the above activity using two different sized syringes.

4. Do you think the syringes will move the same distance this time?

5. Is there any connection between the size of the syringes and the distances they move?

AEROSOLS

➤ Explain the working of Aerosols

Aerosol is a system of particles uniformly distributed in a finely divided state through a gas. Aerosol particles, such as dust, play an important role in the precipitation process, providing the nuclei upon which condensation and freezing take place. Aerosols can be natural or artificial. Examples of natural aerosols are fog, geyser steam. Examples of artificial aerosols are haze, dust, particulate air pollutants and smoke.

Aerosols need no mixing or measuring and are always ready to use at the push of a button. This convenience is rated highly by people to use them in their daily life.

Uses of Aerosols in our daily life:

Aerosols can perform a wide variety of tasks without the need for any additional equipment or tools. Spray paints for example do away with the need for brushes, rollers and turps. Aerosols are ideally suited to:

- Producing fine space sprays such as air fresheners and insect sprays; achieving even coatings such as paints, hair sprays, surface spray insecticides and cooking sprays.
- Producing foams such as carpet shampoo, hair mousse and whipped cream.
- Reaching hard to get places such as with automotive sprays, lubricants and crack and crevice treatments.
- Aerosols can dispense medicines or other products in metered doses down to 50mg.
- Aerosols can jet products long distances for example yard and garden sprays, wasp sprays etc; and One-shot or total discharge valves can empty the contents of a can in one hit making them ideal for fumigation and other similar applications.



7.11 Hair Spray



7.12 Air Freshener



7.13 Inhaler

Principle and Working of Aerosols:

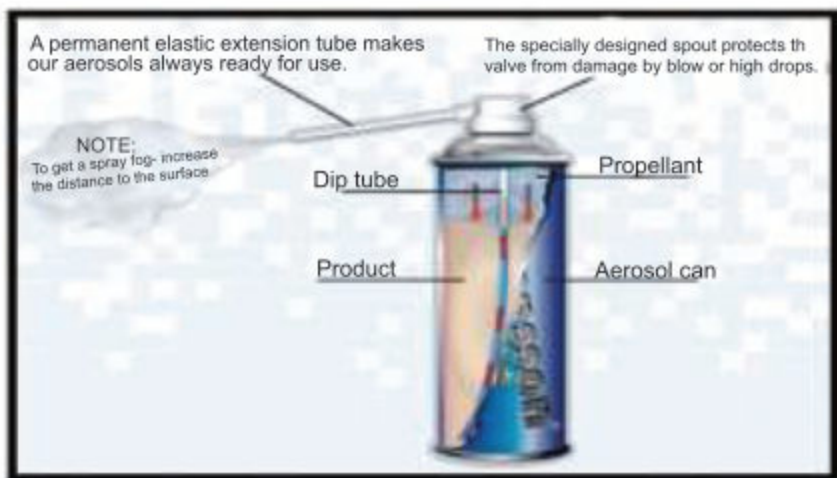
Principle:

A gas under enough pressure will turn into liquid, and when that pressure is relieved, will expand and turn back into a gas. That process is called Vaporization.

Working:

When the valve is open, the pressure on the liquid propellant is instantly reduced. With less pressure, it can begin to boil. Particles break free, forming a gas layer at the top of the can. This pressurized gas layer pushes the liquid product, as well as some of the liquid propellant, up the tube to the nozzle. Some cans, such as spray-paint cans, have a ball bearing inside. If you shake the can, the rattling ball bearing helps to mix up the propellant and the product, so the product is pushed out in a fine mist.

When the liquid flows through the nozzle, the propellant rapidly expands into gas. In some aerosol cans, this action helps to atomize the product, forming an extremely fine spray. In other designs, the evaporating propellant forms bubbles in the product, creating foam.



7.14 Working of Aerosol

ATMOSPHERIC PRESSURE

- Describe the term Atmospheric Pressure

Atmosphere is a thick layer of air surrounds the earth. The air has mass and weight. Atmospheric pressure is defined as the force per unit area exerted against a surface by the weight of the air above that surface. In the diagram below, the pressure at point "X" is shown due to the

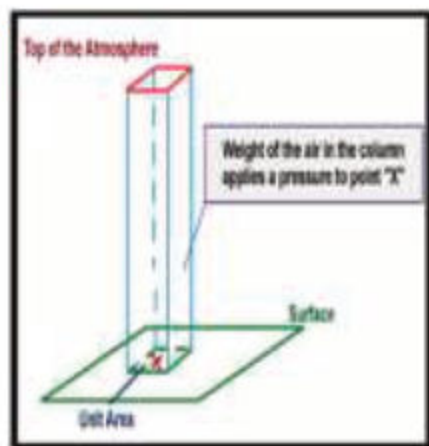


Fig. 7.15 Atmospheric Pressure

weight of the air column above it. At sea level this pressure is equivalent to about 1 Kg on every square centimetre.

Characteristics of Atmospheric Pressure:

1. Atmospheric pressure varies with the height of the object above sea level. At higher altitude, the temperature and density of the air are lower. As a result, the frequency of the collision of molecules is lower. Thus, atmospheric pressure is lower.

2. Atmospheric pressure acts in all directions.

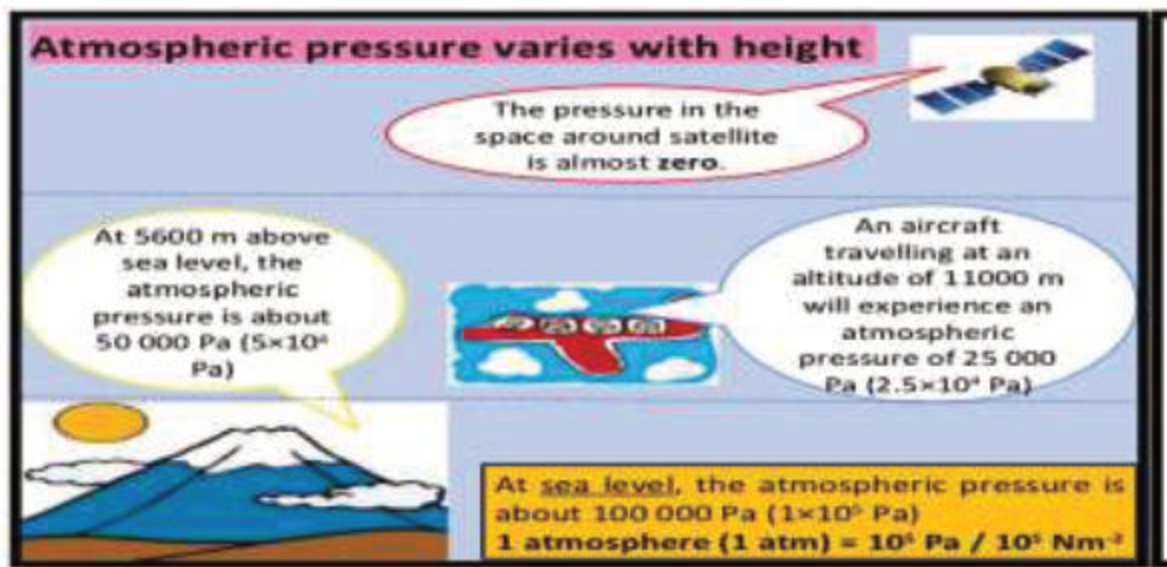


Fig. 7.16 Variation of Pressure with Height

DO YOU KNOW?

Why do your ears pop in airplanes?

As you go up in an airplane, the atmospheric pressure becomes lower than the pressure of the air inside your ears. Your ears pop because they are trying to equalize, or match, the pressure. The same thing happens when the plane is on the way down and your ears have to adjust to a higher atmospheric pressure.

Activity 7.5:

Egg in a Bottle

Materials:

a hard-boiled egg, a glass bottle with a mouth just slightly smaller than the egg, 8-cm by 8-cm (3-inch by 3-inch) piece of newspaper, a match

Set up:

Remove the shell from the egg. Set the egg on the mouth of the bottle to see that the egg does not fit through the mouth.

Procedure:

1. Fold the piece of newspaper into a strip that can be dropped into the bottle, about 1 cm by 8 cm.
2. Light the match and use it to ignite the folded strip of paper. Remove the egg from the mouth of the bottle and drop the burning strip of paper into the bottle.
3. Before the fire goes out, set the egg back onto the mouth of the bottle. Within a few seconds the egg will squeeze through the mouth into the bottle.
4. As it entered the bottle, the egg may have broken into pieces. This happens when the diameter of the egg is more than about 0.5 cm (about $\frac{3}{16}$ inch) larger than the diameter of the bottle's opening. A medium or small egg may be small enough to enter the bottle without breaking.



Questions:

1. Why does the egg slide into the bottle, even though no one is pushing it?

2. What happens when air is heated up?

Measuring Air Pressure

Air pressure can be measured through different devices. The most common device is called the barometer. In a barometer, a column of mercury in a glass tube rises or falls as the weight of the atmosphere changes. Meteorologists describe the atmospheric pressure by how high the mercury rises. An atmosphere (atm) is a unit of measurement equal to the average air pressure at sea level

at a temperature of 15 degrees Celsius. One atmosphere is 760 mm of mercury. Two most common types of barometers are the mercury and the aneroid barometer.

Mercury Barometer

Mercury Barometer is the simplest device to measure atmospheric pressure at a location. It consists of a glass tube closed at one end immersed in a container filled with mercury. Because of the atmospheric pressure mercury rises in the tube as shown.

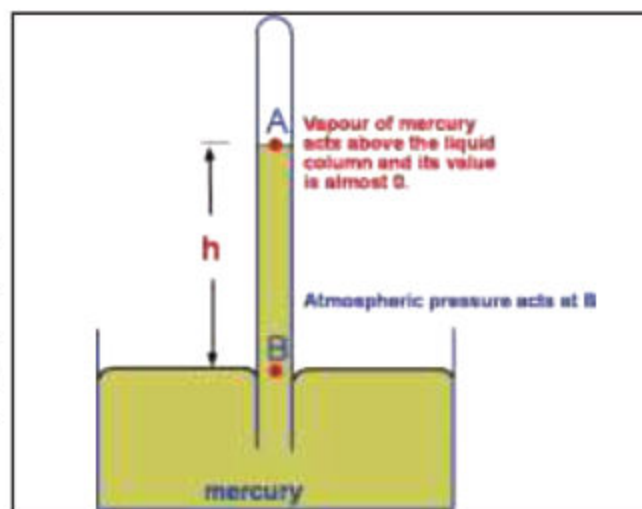


Fig. 7. 17 Mercury Barometer

Aneroid Barometer:

An aneroid barometer, has a sealed, air-tight metal box inside. As the air pressure rises or falls, the box either squashes inward a tiny bit or bends outward. A spring is attached to the box and, as the box moves in and out in response to the changes in air pressure, the spring expands or contracts and moves the pointer on the dial. The dial is calibrated marked with numbers so you can read the air pressure easily. The Aneroid Barometer can be used as an altimeter by mountaineers or in an airplane to determine its altitude.

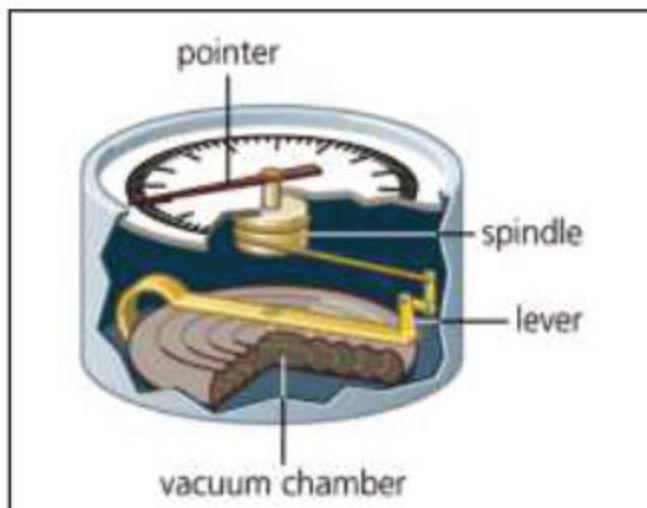


Fig. 7. 18 Aneroid Barometer

SUMMARY

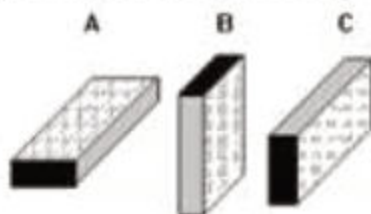
- Pressure is defined as the force per unit area.
- The unit of pressure in the SI system is the pascal (Pa).
- Water pressure is a force that makes a flow of water strong or weak. It varies with the height of the water flow.
- Pascal's Principle states that pressure applied to an enclosed liquid is transmitted equally to every part of the liquid.
- Pascal's Principle is widely used in a Hydraulic System. Hydraulic Brakes, Hydraulic Jack system, Hydraulic Lifts are the commonly used Hydraulic systems.
- The particles in a gas move quickly in all directions. When gas particles hit the walls of their container they cause pressure.
- Pneumatic technology deals with the study of behaviour and applications of compressed air.
- Pneumatic systems are used in Spray guns, Pumps, dentistry tools etc.
- Pneumatic system uses air that is compressed in order to transmit and control energy.
- An aerosol is a suspension of fine solid particles or liquid droplets, in air or another gas.
- Atmospheric pressure is around us all the time.
- The atmospheric pressure at sea level has a mean value of 101,325 pascals (roughly 14.7 pounds per square inch).
- A barometer is a scientific instrument used to measure atmospheric pressure.

EXERCISE

Q1. Choose the correct answer:

- I. Force has
 - a. Magnitude
 - b. Direction
 - c. Magnitude and direction both
 - d. None of these
- II. What is the unit of pressure in SI units:
 - a. Joule
 - b. Kg/m^2
 - c. Newton
 - d. Pascal
- III. Pressure increases when:
 - a. Area of contact is increased
 - b. Area of contact is decreased
 - c. Magnitude of force is decreased
 - d. Area of contact remain same
- IV. The pressure in a liquid is transmitted equally in all directions, this principle is called:
 - a. Archimedes' Principle
 - b. Pascal's Principle
 - c. Bernoulli's Principle
 - d. Le Chateliers's Principle
- V. Atmospheric Pressure is calculated by the:
 - a. Height of water column in barometer
 - b. Height of mercury column in barometer
 - c. Height of lime column in barometer
 - d. Height of oil column in barometer
- VI. Amount of pressure of liquid increases with increase of :
 - a. Volume
 - b. Depth
 - c. Base area
 - d. Mass
- VII. We may feel internal injury or altitude sickness at higher or lower altitudes than sea level because:
 - a. The pressure of our body remains the same
 - b. Our body adjusts the pressure slowly
 - c. Our body adjust the pressure quickly
 - d. Our body unable to adjust the pressure

VIII. A rectangular piece of wood is kept in 3 different ways on a table. Pressure exerted by the wood on the table will be:



- a. Maximum in position A
- b. Maximum in position B
- c. Maximum in position C
- d. Equal in all cases

IX. Which fluid is used in hydraulic power systems?

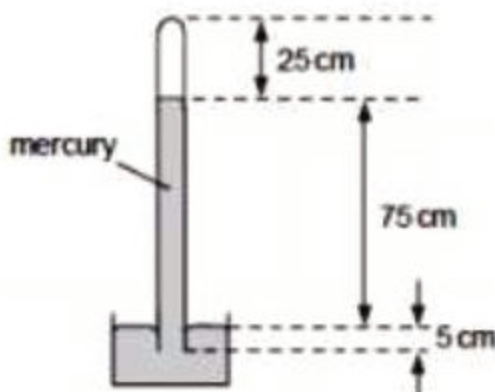
- a. Water
- b. Oil
- c. Non-compressible fluid
- d. All of these

X. Pressure in gases is due to:

- a. Collision of molecules with walls of container
- b. The atmosphere
- c. Collision of molecules on base of container
- d. Collisions of molecules with each other

XI. The diagram shows a mercury barometer. Which distance is used to calculate pressure of the atmosphere?

- a. 25 cm
- b. 75 cm
- c. 80 cm
- d. 100 cm



Q2. A large aquarium for fish is filled with water. The weight of the aquarium with the water in it is 10000N. The base of the aquarium has an area of 1.6m^2 . Calculate the pressure exerted on the base of the aquarium by the water.

Answer: $P=6250\text{Pa}$

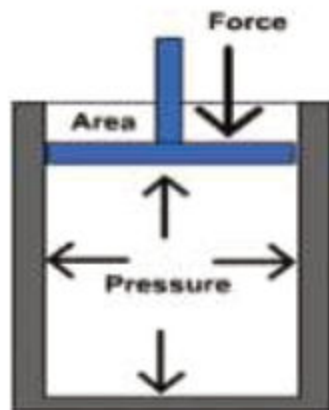
Q3. Imagine the atmospheric pressure at present is 101200 Pa . You are holding your hand, and the atmosphere is exerting a force of its own on your hand. If the area of your palm is 0.006 m^2 , what is the force exerted on your hand.

Answer: $F=607 \text{ N}$

Q4. Look at the diagram to the right to answer this question:

If a force of 46 N was applied to a system as you pushed it down and the area of piston is 0.5 m^2 , then what would be the pressure inside the cylinder?

Answer: 92 N/m^2



Q5. Define Aerosols and explain the working of Aerosols.

Q6. Why do camels have such large feet?

Q7. A force of 300 N creates a pressure of 4 N/m^2 . Over what area is the force acting?

Answer: 75 m^2

Q8. State Pascal's Law and explain an application of this law.

Project

Construct a simple model barometer

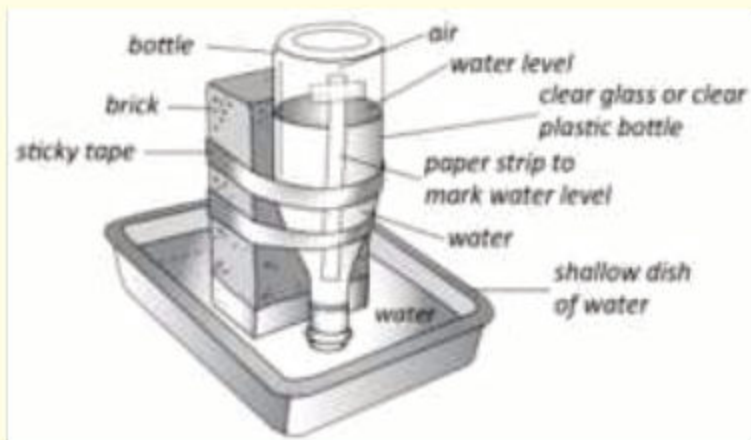
This model barometer is based on the variation of air pressure by changing the height of a column of water.

Materials needed:

1 long narrow clear-glass bottle, a paper strip to mark the water level, a brick, sticky tape, a shallow dish and water.

Procedure:

- First fill the glass bottle with water.
- Hold the dish over the top of the bottle and turn the bottle upside down quickly and carefully.
- The bottle should be about one-third filled with air. If necessary, tilt the bottle slightly to let some more air in.
- Tape the bottle to a brick so that it will not fall over and stick a strip of paper on the bottle and mark the level of the water at the start of work.



- Look at the barometer at the same time every day for a month. Each day write down the date, time, the weather, and whether the water level in the barometer is going up or going down (compared to the previous day). Remember, when the air pressure increases, the level of the water in the bottle will rise. When the air pressure falls, the level of the water in the bottle will fall. At the end of the month, look at your results and discuss your observations with your teacher.